

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech (CL/ME) Examination

March 2018

CL 202 / 202.01 – Strength of Materials

Date: 28.03.2018, Wednesday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Define the following terms:

[07]

- I. In a simply supported beam, bending moment at the end
 - a) Is always zero if it does not carry couple at the end
 - b) Is zero, if the beam has uniformly distributed load only
 - c) Is zero if the beam has concentrated loads only
 - d) May or may not be zero
- II. In a simply supported beam subjected to uniformly distributed load (w) over the entire length (l), maximum Bending moment is
 - a) $wl^2/8$ at the mid-point
 - b) $wl^2/8$ at the end
 - c) $wl^2/4$
 - d) $wl/2$
- III. The moment of inertia of a body is always minimum with respect to its
 - a) Base
 - b) Centroidal axis
 - c) Vertical axis
 - d) Horizontal axis
- IV. Shear stress is zero at the
 - a) Outermost fiber
 - b) Central fiber
 - c) Neither outermost nor central fiber
 - d) None
- V. In a simply supported beam loaded with U.D.L over the whole section, the bending stress isat top andat bottom.

- a) Compressive, tensile
 - b) Tensile, compressive
 - c) Tensile, zero
 - d) Compressive, zero
- VI. For any part of the beam, between two concentrated load Shear force diagram is a
- a) Horizontal straight line
 - b) Vertical straight line
 - c) Line inclined to x-axis
 - d) Parabola
- VII. When a shaft is subjected to a twisting moment, every cross-section of the shaft will be under
- a) Tensile stress
 - b) Compressive stress
 - c) Shear stress
 - d) All of these

Q - 2 (A) Determine the moments of inertia and the radius of gyration of the shaded area with respect to the centroidal axis as shown in Figure 1. [07]

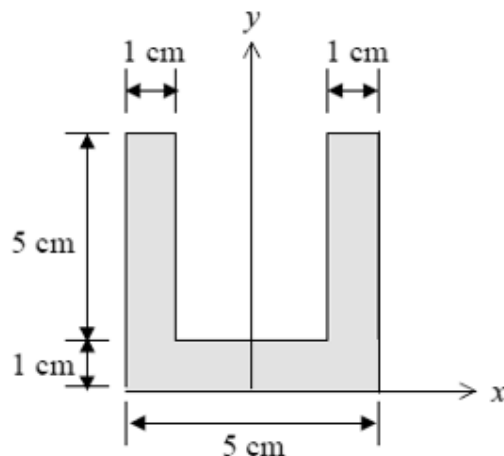


Figure 1

OR

- (A) Find diameter of a solid cylindrical shaft is to transmits 300kW power at 100 r.p.m if shear stress is not to exceed 80 N/mm^2 . [07]
- (B) Determine the moment of inertia of the I-section about x-x axis passing through the C.G. of cross section as shown in Figure 2. [07]

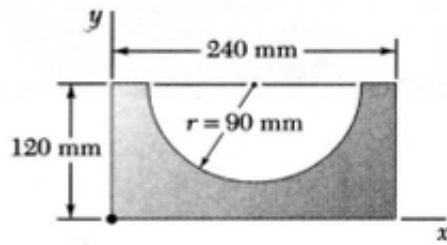


Figure 2

OR

- (B) A hollow circular shaft of external & internal diameter of 20 cm & 10 cm [07]
respectively, the shear stress in the shaft is not exceeds 40 N/mm^2 . Find the maximum
torque which the shaft can safely transmit.

- Q - 3 (A) Determine the moment of inertia of rectangular section about the axis X-X passing [04]
through its C.G. and parallel to its width. The width of rectangular section is B and
depth of rectangular section is D.

- (B) (i) Draw Shear Force & Bending Moment diagram of beam as shown in Figure 3. [10]
(ii) Also draw the shear force & bending moment diagram if $P = 20 \text{ kN}$ and $L = 6 \text{ m}$.

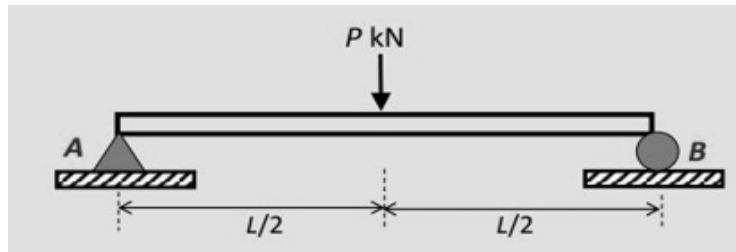


Figure 3

OR

- (B) Draw Shear Force & Bending Moment diagram of beam as shown in Figure 4 (a) & [10]
(b).

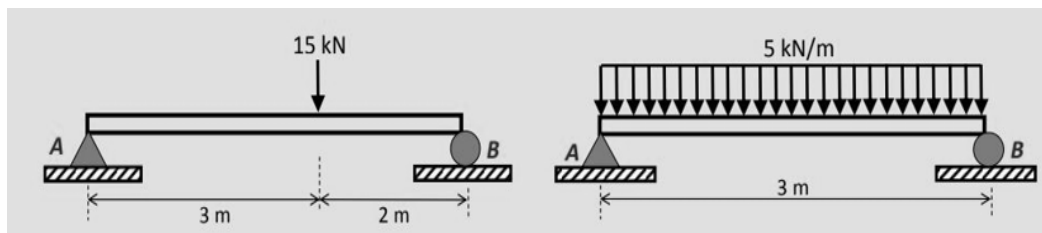


Figure 4 (a)

Figure 4 (b)

SECTION II

Q - 4 (A) Define following terms: **(Any 3)** [03]

(i). Shear Force (ii). Point of Contraflexure (iii). Torsion (iv). Bending Moment

(B) The tensile stresses at a point across two mutually perpendicular planes are 85 N/mm^2 [04]
and 25 N/mm^2 and shear stress is 60 N/mm^2 . Calculate the major principal stress,
minor principal stress and obliquity of principal plane.

Q - 5 (A) At a point within a body subjected to two mutually perpendicular directions, the [06]
stresses are 30 MPa compressive and 15 MPa tensile. Each of the above stresses is
accompanied by shear stress of 10 MPa . Find the normal and shear stress making the
angle 30° to plane of first stress as well as resultant stress & its obliquity.

(B) A rectangular beam 300 mm deep is simply supported over a span of 4 m . Determine [08]
uniformly distributed load which the beam may carry, if the bending stress should not
exceed 120 N/mm^2 . Take $I = 8 \times 10^6 \text{ mm}^4$.

OR

(B) A simply supported wooden beam of span 1.3 m having a cross section 150 mm wide [08]
by 250 mm deep carries a point load **W** at the center. The permissible stresses are 7
 N/mm^2 is bending and 1 N/mm^2 is shearing. Calculate the safe load **W**.

Q - 6 (A) Draw the mohr's circle when a body is subjected to two mutually perpendicular [06]
principal tensile stresses of unequal intensities.

(B) A circular beam 100 mm diameter is subjected to maximum shear force of 5 kN . [08]
Calculate,
i. Average shear stress,
ii. Maximum shear stress,
iii. Shear stress at a distance of 40 mm above N.A.

OR

(B) A cantilever beam of length 2 m fails when a load of 2 kN is applied at the free end. If [08]
the section of beam is $40 \text{ mm} \times 60 \text{ mm}$, find the stress at the failure.
